

INTERMEDIATE CASING CEMENTING EVALUATION

SYNTHETIC DEMONSTRATION DOCUMENT - NOT REAL OPERATIONAL DATA (SIH PROTOTYPE)

WELL IDENTIFIER: NWIS-W005
FORMATION: Demo-Girujan Clay
EVENT TYPE: CEMENTING_ISSUE

OPERATIONAL DATE: 2022-04-10
MEASURED DEPTH: 1620.0 m
SEVERITY LEVEL: HIGH

OPERATIONAL DESCRIPTION

Ran 9-5/8 inch intermediate casing to 1620m across Demo-Girujan Clay interval.
Pumped 450 sacks of Class G cement slurry followed by 1.25 SG displacement mud.
Cement top found lower than calculated; post-job Cement Bond Log (CBL) revealed channeling.

ROOT CAUSE ANALYSIS

Inadequate mud displacement efficiency and gas channeling through setting cement slurry.
Washouts in soft Girujan clay sections prevented laminar slurry placement.

REMEDIAL MITIGATION

- 1. Perforated 9-5/8 inch casing at 1580m using 4-inch casing guns (4 SPF).
- 2. Established pump-in injection rate and performed remedial cement squeeze with micro-fine cement.
- 3. Pressure tested squeeze interval to 2500 psi for 15 minutes; successfully sealed annulus.